Exercise 4:

(b)

Intercept: The predicted value when Price = 0, the store is not located in an Urban area (Urban = No), and is not in the United States (US = No).

Price: The negative coefficient indicates that for every 1 unit increase in price (e.g., $1), the sales will decrease by an amount equal to this coefficient (assuming other factors remain constant).

Urban[Yes]: This is a dummy variable. If the store is located in an Urban area (Urban == Yes), the sales will differ by an amount equal to this coefficient compared to Rural (No). (Note: Typically, the p-value of this variable is large, indicating that the difference is not statistically significant).

US[Yes]: If the store is located in the United States (US == Yes), the average sales will be higher by an amount equal to this coefficient compared to stores not located in the US (US == No).

(c)

$$Sales = \beta_0 + \beta_1 \times Price + \beta_2 \times x_{Urban[Yes]} + \beta_3 \times x_{US[Yes]}$$

$x_{Urban[Yes]} = 1$ if it is Urban (and 0 elsewhere), similar to $x_{US[Yes]}$.

(d)

Since P>|t| of Urban(Yes) > α = 0.05, and others < α = 0.05, I remove Urban(Yes) and use others for fit_smaller

Exercise 6

(a)

- The population model is:

$$y = 2 + 2x_1 + 0.3x_2 + \varepsilon$$

- The corresponding population coefficients are:
  - $\beta_0 = 2$ (intercept)
  - $\beta_1 = 2$ (coefficient of x_1)
  - $\beta_2 = 0.3$ (coefficient of x_2)

(b)

- The correlation coefficient between x_1 and x_2 is expected to be high (approximately 0.85 or higher) because x_2 is constructed based on x_1. The scatter plot should show points following an upward-sloping linear pattern.

(c)

- When both x_1 and x_2 are included in the same regression model, the p-value of one of the variables (often x_2) may become greater than 0.05, indicating that the variable is not statistically significant. This means that we fail to reject the null hypothesis
  H_0: $\beta_2 = 0$
  This may occur even though the true population coefficient of x_2 is 0.3.
- The Variance Inflation Factor (VIF) values may be very high (often exceeding 5 or 10), indicating severe multicollinearity that inflates the standard errors of the estimated coefficients.

(d)

- When each predictor is fitted separately (regressing y on x_1 alone or y on x_2 alone), both variables may show statistically significant relationships with y ($p < 0.05$).

- This result does not contradict the result in (c). Because x_1 and x_2 are highly correlated, including both in the same model makes it difficult to distinguish their individual contributions to explaining y. This can substantially increase the variances and standard errors of the estimated coefficients, causing one variable to lose statistical significance.

Exercise 7

(a)

Model 1 ($R^2 = 0.3439, RSE = 62.5151$)

Model 2 ($R^2 = 0.4801, RSE = 55.7782$)

Model 3 ($R^2 = 0.5177$, $RSE = 54.1542$)

(b) Examine the p-value from `anova_lm` is

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	438.0	1.362709e+06	0.0	NaN	NaN	NaN
1	431.0	1.263986e+06	7.0	98722.908072	4.809002	0.000031

The p-value is 0.000031 less than 0.05. Which means we fail to reject H0, we conclude that the full model provides a statistically significant improvement over the reduced model.

(c) When examining the VIF table for the full model, some serum variables (particularly s1 and s2, or possibly s3 and s4) may have high VIF values (often greater than 10). This indicates that these serum measurements are strongly correlated with other predictors in the model, suggesting severe multicollinearity.

(d) Recommendation: Consider bmi, s5, and bp as candidate predictors, based on their t-statistics and p-values in the full model summary (m3.summary()). The final selection should be confirmed using the actual regression results.

Explanation: The bmi variable is generally strongly associated with diabetes disease progression. The serum variable s5, which is related to a transformed serum triglyceride measurement in the diabetes dataset, may also be an important predictor. However, because of multicollinearity among the serum measurements (s1–s6), as discussed in Part (c), their estimated regression coefficients may be unstable. Therefore, physicians should consider both statistical evidence and clinical relevance when interpreting the results.